

L06 — Searching: Linear Search and Binary Search with Complexity Analysis

Unit: 1 | Chapter: Fundamentals of Data Structures & Arrays | BT Level: BT4 | CO: CO1, CO5

Learning Objectives

- Implement linear search and analyze its complexity
 - Implement binary search (iterative and recursive) and analyze its complexity
 - State the precondition for binary search
 - Compare both algorithms and choose appropriately
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1. Introduction to Searching

Searching is the process of finding the location of a target element within a data structure.

Two fundamental search algorithms on arrays:

1. **Linear Search** — works on any array (sorted or unsorted)
 2. **Binary Search** — works only on **sorted** arrays, dramatically faster
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2. Linear Search

2.1 Algorithm

Scan each element one by one from left to right until the target is found or the array is exhausted.

```
def linear_search(arr, n, target):
    for i in range(n):
        if arr[i] == target:
            return i          # Found at index i
    return -1                 # Not found
```

2.2 Trace Example

Array: [45, 12, 78, 3, 56, 21], Target: 56

Step	i	arr[i]	Match?
1	0	45	No
2	1	12	No
3	2	78	No
4	3	3	No

5 4 56 Yes → return 4

2.3 Complexity Analysis

Case	Condition	Comparisons	Time
Best	Target at index 0	1	O(1)
Worst	Target at last index or absent n	n	O(n)
Average	Target at random position	n/2	O(n)

Space: O(1) — no extra memory

2.4 Improved Linear Search: Sentinel Search

Avoids checking the loop boundary on every iteration:

```
def sentinel_search(arr, n, target):
    last = arr[n - 1]
    arr[n - 1] = target      # Place sentinel at end
    i = 0
    while arr[i] != target:
        i += 1
    arr[n - 1] = last        # Restore
    if i < n - 1 or arr[n - 1] == target:
        return i
    return -1
```

Reduces the number of comparisons per iteration from 2 to 1.

3. Binary Search

3.1 Precondition

The array MUST be sorted (ascending or descending).

Binary search exploits the sorted order to **eliminate half the search space** in each step.

3.2 Core Idea

Maintain three pointers: low, mid, high.

- If `arr[mid] == target`: found!
- If `arr[mid] < target`: target must be in right half → `low = mid + 1`
- If `arr[mid] > target`: target must be in left half → `high = mid - 1`

3.3 Iterative Binary Search

```
def binary_search_iterative(arr, n, target):
    low = 0
    high = n - 1
```

```

while low <= high:
    mid = low + (high - low) // 2    # Avoids integer overflow vs (low + high) // 2

    if arr[mid] == target:
        return mid                # Found
    elif arr[mid] < target:
        low = mid + 1              # Search right half
    else:
        high = mid - 1             # Search left half

return -1                          # Not found

```

Note: Use `mid = low + (high - low) // 2` instead of `(low + high) // 2` to prevent integer overflow in languages with fixed-width integers.

3.4 Recursive Binary Search

```

def binary_search_recursive(arr, low, high, target):
    if low > high:
        return -1                # Base case: not found

    mid = low + (high - low) // 2

    if arr[mid] == target:
        return mid
    elif arr[mid] < target:
        return binary_search_recursive(arr, mid + 1, high, target)
    else:
        return binary_search_recursive(arr, low, mid - 1, target)

```

3.5 Trace Example

Array: [3, 12, 21, 45, 56, 78], Target: 45, n = 6

Iteration	low	high	mid	arr[mid]	Decision
1	0	5	2	21	21 < 45 → right half
2	3	5	4	56	56 > 45 → left half
3	3	3	3	45	Found! Return 3

Only 3 comparisons to find 45 in 6 elements. Linear search would take 4.

3.6 Complexity Analysis

Each iteration halves the search space:

After k iterations, search space = $n / 2^k$

Algorithm terminates when search space = 1: $n / 2^k = 1 \implies k = \log_2(n)$

Case	Comparisons	Time
Best	1	$O(1)$

Worst $\lfloor \log_2(n) \rfloor + 1$ $O(\log n)$
Average $O(\log n)$ $O(\log n)$

Space:

- Iterative: $O(1)$
 - Recursive: $O(\log n)$ — call stack depth
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4. Comparison: Linear vs Binary Search

Property	Linear Search	Binary Search
Precondition	None	Array must be sorted
Best case	$O(1)$	$O(1)$
Worst case	$O(n)$	$O(\log n)$
Average case	$O(n)$	$O(\log n)$
Space (iterative)	$O(1)$	$O(1)$
Works on linked list	Yes	No (no random access)
Implementation	Simple	Moderate

When to Use Each

Situation	Choose
Unsorted data	Linear Search
Sorted data, frequent queries	Binary Search
Small array ($n < 50$)	Either (difference negligible)
Single search in sorted data	Linear (avoids sort overhead)

5. Performance at Scale

For $n = 1,000,000$:

- Linear search: up to **1,000,000** comparisons
 - Binary search: at most **20** comparisons ($\log_2(10^6) \approx 20$)
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6. Variants of Binary Search

Find First Occurrence

```
def first_occurrence(arr, n, target):  
    low, high, result = 0, n - 1, -1  
    while low <= high:  
        mid = low + (high - low) // 2  
        if arr[mid] == target:  
            result = mid
```

```
        high = mid - 1    # Continue searching left
    elif arr[mid] < target:
        low = mid + 1
    else:
        high = mid - 1
return result
```

Find Last Occurrence

```
def last_occurrence(arr, n, target):
    low, high, result = 0, n - 1, -1
    while low <= high:
        mid = low + (high - low) // 2
        if arr[mid] == target:
            result = mid
            low = mid + 1    # Continue searching right
        elif arr[mid] < target:
            low = mid + 1
        else:
            high = mid - 1
    return result
```

Summary

- **Linear search:** Simple, $O(n)$ worst case, works on any array.
- **Binary search:** Requires sorted array, $O(\log n)$ worst case — drastically faster for large n .
- Binary search eliminates half the search space per step by leveraging sorted order.
- For $n = 10^6$, binary search takes ~ 20 steps vs up to 10^6 for linear search.

References

- Cormen et al., *Introduction to Algorithms*, Ch. 2.1 (MIT Press, 2022)
- Skiena, *The Algorithm Design Manual*, Ch. 4.9 (Springer, 2020)
- Laaksonen, *Guide to Competitive Programming*, Ch. 3 (Springer, 2024)